

Fine-Grained Tai Chi Action Recognition from Skeleton Data: A Comparative Study of BiLSTM and STGCN Architectures

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Abstract

Tai Chi is a traditional martial art from China with slow and smooth movements, which look alike for different styles, making automated recognition an extremely fine-grained classification problem. Current solutions either view the skeleton joints as independent features without spatial body structure information or use raw videos which are expensive in terms of computation. We address this by framing the task as a 10-class action recognition problem on 3D skeleton data from the HIT605 TaiChi dataset. We explore three different architectures: a standard BiLSTM model, a BiLSTM with temporal attention mechanism, and a Spatio-Temporal Graph Convolutional Networks (STGCN), which captures the structural topology of human body. Our experiments prove that the BiLSTM baseline has achieved the best Top-1 accuracy at 89.25% and Macro F1 score at 0.8825. The STGCN has achieved comparable performance at 86.40% with only 23,514 parameters, 25 times less than the BiLSTM models, showing highly efficient model size. Temporal attention is proven to have no effect on performance in our ablation study due to the sufficient local context in the BiLSTM final hidden state for 64 frames.

Introduction

Tai Chi (Taijiquan) is a martial art practiced worldwide for health and wellness [1]. Automated recognition of Tai Chi movements has many potential applications, including sports training and physiotherapy. However, due to the precise nature of Tai Chi, with various movements requiring slow and controlled motion, classification becomes particularly difficult.

The challenge is to discern the differences among visually identical body positions that vary only in terms of slightly different joint postures. For instance, *Single Whip* and *Lift up Hand* share similar body postures above the waist but differ in terms of weight distribution and foot positions.

Previous studies have included using trajectory features with SVMs [2], motion capture information for gesture recognition [3], and video-based I3D models for quality assessment [1]. However, no study has directly compared time series and graph-based spatial models using the same skeleton dataset.

Our three contributions are: (1) adaptation and comparison of BiLSTM and STGCN architectures on a task of 10-class Tai Chi recognition; (2) an extensive ablation study that includes temporal attention analysis; and (3) our own 17-node human skeleton graph for STGCN on the HIT605 dataset.

Related Work

Application Perspective. Xu et al. [2] use trajectory primitives extracted from skeleton data and apply SVM to classify Tai Chi actions, thus setting a benchmark based on traditional machine learning algorithms. Wang et al. [1] introduce a video-based approach for assessing the quality of action using I3D features. Tits et al. [3] create a multimodal motion capture dataset for analyzing gestures in Taijiquan. Scott et al. [6] create the PSU-TMM100 dataset for stability estimation based on human pose.

Methodological Perspective. BiLSTM techniques have been extensively applied in the domain of skeleton-based action recognition by modeling joint locations as sequential feature vectors [5]. STGCN [4] applies graph convolution on human body skeletons using an adjacency matrix that captures joint connectivity. The technique is considered as the baseline method for skeleton-based action recognition. Our contribution is the direct comparison of these two methodologies on Tai Chi data.

Methodology

Data Representation

Each skeletal frame consists of $N = 17$ joint coordinates in 3D space, resulting in a feature tensor $\mathbf{X} \in R^{17 \times 3 \times T}$ with T being the number of frames. A sliding window of size $W = 64$ frames and stride 32 is applied to create fixed-length windows that are labelled according to a majority vote from frame labels. We get 243,461 windows for training and 3,508 for testing.

Model 1: BiLSTM Baseline

The skeleton tensor is flattened into $\mathbf{x} \in R^{T \times 51}$ ($51 = 17 \text{ joints} \times 3 \text{ coordinates}$). A two-layer BiLSTM with hidden size 128

is used for processing the input sequence:

$$\vec{\mathbf{h}}_t, \overleftarrow{\mathbf{h}}_t = \text{BiLSTM}(\mathbf{x}_t) \quad (1)$$

The concatenation of final hidden layers is then passed through dropout and a linear classifier with softmax activation.

Model 2: BiLSTM + Temporal Attention

The BiLSTM is further enhanced by adding temporal attention as follows:

$$\alpha_t = \text{softmax}(\mathbf{W}_a \mathbf{h}_t), \quad \mathbf{c} = \sum_t \alpha_t \mathbf{h}_t \quad (2)$$

This makes the model pay attention to the most discriminative frames rather than depending only on the last hidden state, and thus solves the bottleneck issue of sequence modeling.

Model 3: STGCN

In the STGCN model, the skeleton is considered a spatial graph $\mathcal{G} = (V, E)$ with V as 17 nodes and E containing 18 anatomical edges (Figure 1). Normalisation of adjacency is done as follows:

$$\hat{A} = \tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}}, \quad \tilde{A} = A + I \quad (3)$$

An ST-Block in the STGCN model is formed by first applying TCN followed by GCN:

$$H^{(l+1)} = \sigma(\hat{A} H^{(l)} \Theta^{(l)}) \quad (4)$$

Input data is first resized to $(B, 3, 17, T)$ and then processed by two consecutive ST-Blocks having $(3 \rightarrow 64 \rightarrow 128)$ channels. Global average pooling and a linear layer make up the head.

All models use cross-entropy loss with Adam optimiser ($\text{lr}=10^{-3}$) using step learning rate decay for 50 epochs.

17-Joint Skeleton Graph (STGCN Adjacency)

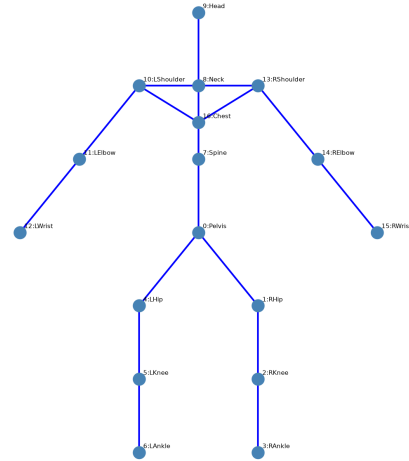


Figure 1: Graph structure of 17 body joints as the adjacency matrix in STGCN. Each circle represents a body joint and each line represents the anatomical relation between joints (e.g., shoulder→elbow, knee→ankle). This graph is normalised as $\hat{A} = \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}$ before being passed to the graph convolution layer.

Experiments

Dataset

The dataset we used is HIT605 TaiChi [2], which contains 1,040 skeleton sequences (1,024 training, 16 testing) for 10 Yang-style Tai Chi movements captured by a motion capture system at 30 fps.

Data Preprocessing

Skeleton sequences with shapes of $(17, 3, T)$ are divided into windows of length 64 with strides of 32, each window being assigned the majority class label. In the case of STGCN, the sequences are transposed to have shapes of $(3, 17, T)$ to be consistent with the (C, N, T) input format of the model.

The dataset exhibits significant class imbalance (Figure 2): *Grasp Bird’s Tail* consists of 1,689,088 samples whereas *Hold the Lute* is comprised of just 442,368 samples – a $3.8\times$ difference. To mitigate the class imbalance, we employ a standard CrossEntropyLoss function without class weighting, and use Macro F1 as our main performance measure.

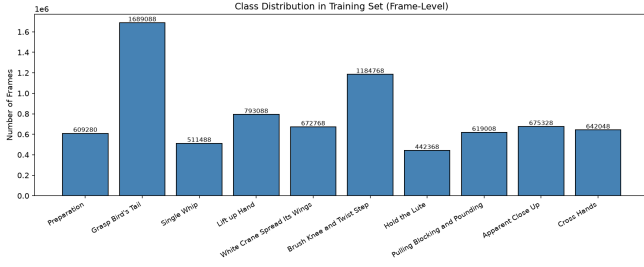


Figure 2: Class distribution for each frame in the HIT605 training set. *Grasp Bird's Tail* (1,689,088 frames) is the most common class, whereas *Hold the Lute* (442,368 frames) is the least common class, with a difference ratio of 3.8. This ratio explains the low performance for the *Hold the Lute* class in terms of the F1 score (0.77).

Evaluation Metrics

Top-1 Accuracy and Macro F1-score are used as primary metrics. Macro F1 is chosen because it weighs all classes equally irrespective of their frequencies, which is a more appropriate metric when dealing with imbalanced data. Per-class precision, recall and F1-scores are provided and visualized using confusion matrices.

Main Results

Table 1: Ablation study results on HIT605 TaiChi test set.

Model	Acc	Macro F1	Params
BiLSTM (baseline)	0.8925	0.8825	583K
BiLSTM + Attention	0.8863	0.8702	583K
STGCN	0.8640	0.8506	23K

Table 1 below summarises our ablation study. The baseline BiLSTM model reaches maximum accuracy (89.25%) and Macro F1 (0.8825). Addition of temporal attention had no impact on performance. This could be explained by the fact that a 64-frame window gives enough local context to the BiLSTM hidden states at each timestamp, making temporal attention irrelevant in this task setting.

STGCN achieves accuracy of 86.40% with only 23,514 parameters, which is $25\times$ less compared to BiLSTM models. This proves that explicit modeling of spatial joint topology is very parameter-efficient. The lower accuracy in comparison with BiLSTM is probably because there is not enough data (1,024 sequences) to make use of graph topology effectively – a known limitation of GCN-based methods.

Visualisations and Analysis

Confusion matrices. Figures 4 and 5 show the confusion matrices of BiLSTM+Attention and STGCN correspondingly.

BiLSTM has most of the mistakes between the temporally adjacent actions: the actions of *Single Whip* are mistaken for *Lift up Hand* (36 times), and *Brush Knee and Twist Step* for *Hold the Lute* (45 times). STGCN has misclassification of *Lift up Hand* to *Preparation* (39 times) because of the similar upper body spatial topology – a unique mistake pattern indicating that the two algorithms recognize different discriminative features. The best performing class in both cases is *Grasp Bird's Tail* (recall = 0.99).

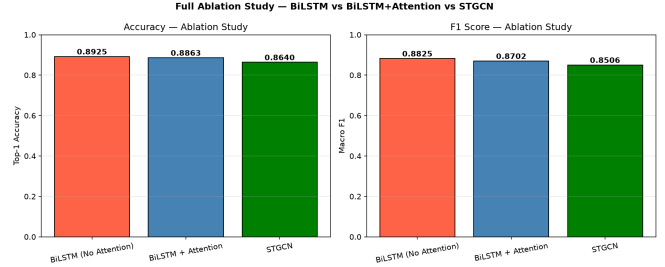


Figure 3: Complete ablation experiment for all three architectures. **Left:** Top-1 Accuracy. **Right:** Macro F1 Score. The BiLSTM (red) obtains the best performance in both metrics. The BiLSTM+Attention (blue) is second best, proving that attention does not contribute any improvements at this window size. STGCN (green) is comparable despite having $25\times$ fewer parameters.

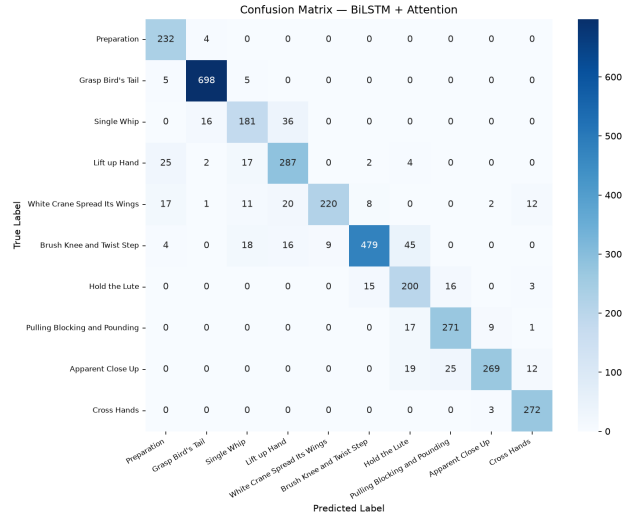


Figure 4: Confusion Matrix for BiLSTM+Attention. Rows are representations of true labels; columns are representations of predicted labels. The prominent diagonal proves the high accuracy per class. The main mistakes appear between temporally adjacent actions: *Single Whip* is mistaken for *Lift up Hand* (36 cases), *Brush Knee* is mistaken for *Hold the Lute* (45 cases), indicating that the model confuses actions with similar motion trajectories.

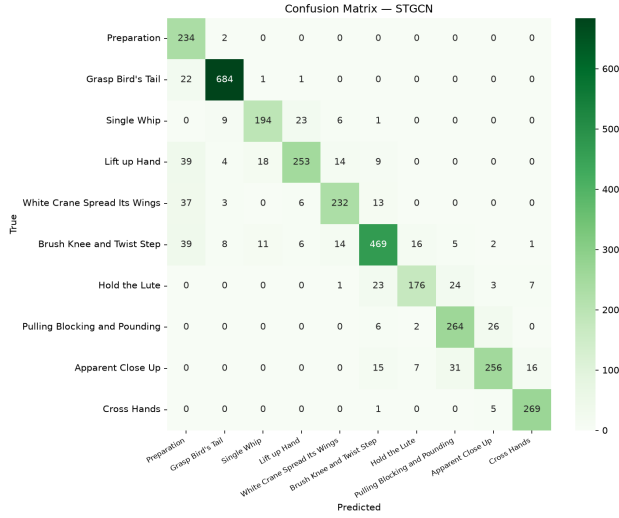


Figure 5: Confusion matrix for STGCN model. Unlike BiLSTM, the main error type here is spatial and not temporal. *Lift up Hand* is mistaken as *Preparation* in 39 cases due to common upper body joint positioning in these actions.

Training curves. Figure 6 illustrates the graphs for training loss and test accuracy for all models. Both BiLSTM models demonstrate a similar convergence, which proves that there is no need for the attention mechanism at this window size. STGCN converges slower but manages to reach a competitive accuracy plateau, which proves that the graph structure needs more time to learn the spatial dependency.

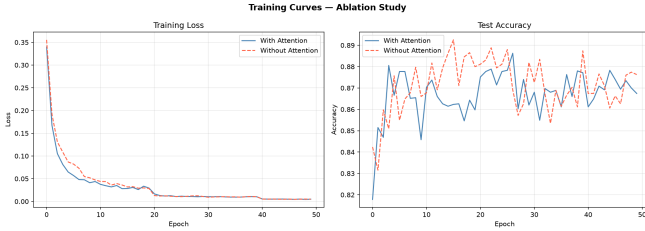


Figure 6: Training loss and test accuracy for BiLSTM model (red dashed line) vs BiLSTM with attention model (blue solid line). These models converge to the same loss and test accuracy, which proves that there is no benefit of the attention mechanism at this window size. It can be seen that a large gap between training accuracy ($\approx 99.8\%$) and test accuracy ($\approx 89\%$) starts from epoch 5.

Attention weights. Attention weights shown in Figure 7 reach their maximum value at a later time point in each window, indicating that end points are more discriminating than start points, which is intuitive for Tai Chi poses since end poses are more distinguishing between different classes.

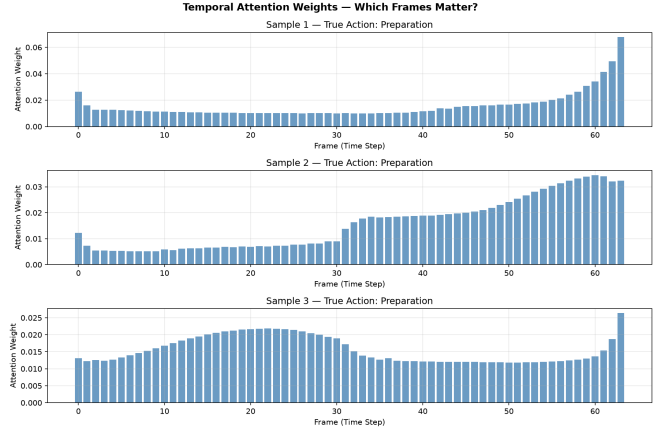


Figure 7: Temporal attention weights of three test samples (all labeled *Preparation*). On the horizontal axis is the frame index (0–64) of the window; on the vertical axis is the attention weight assigned to each frame. For all three samples, attention weights increase with the frames’ progression, signifying that endpoints of movement are more discriminative than its onset – a logical result since the last Tai Chi position is more distinguishable than the first one.

Discussion

All three architectures are heavily overfitted: training accuracy hits 99.8% but test accuracy stagnates at 86–89%. This is an acknowledged problem with LSTM models applied to dense frame-level skeleton sequences. We tried dropout values of 0.3 and 0.5, with 0.3 being best for both BiLSTM versions. However, despite that, there is a huge gap between training and test accuracy, indicating that due to highly correlated training samples generated by the sliding window with 50% overlap, the model overfits the data. In future research, we have to try L2 regularization, data augmentation through joint coordinate jittering and temporal warping, and test with non-overlapping windows to receive unbiased sequence-level predictions. Moreover, the small amount of testing data (16 sequences, 3,508 windows) is the source of metrics variance – one wrongly predicted sequence may significantly change the accuracy value.

It should be noted that the architectural advantage that STGCN enjoys does not lead to superior results, and the performance of BiLSTM is higher, which indicates that the problem here is the size of the dataset. With only 1,024 training samples, the inductive bias of BiLSTM (temporal sequence modelling) generalises better.

Conclusion

In this research, we compared BiLSTM and STGCN architectures for fine-grained Tai Chi action recognition. Our BiLSTM has generated 89.25% Top-1 results on HIT605, and 86.40% with STGCN while only applying $25\times$ smaller number of pa-

rameters. Our study reveals that while using 64 frames, temporal attention does not improve results, while modelling spatial relations on graphs is very beneficial when there is enough training data. Future research will involve cross-dataset assessment with UMONS-TAICHI, a learnable adjacency matrix of STGCN, and techniques aimed at preventing overfitting.

References

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